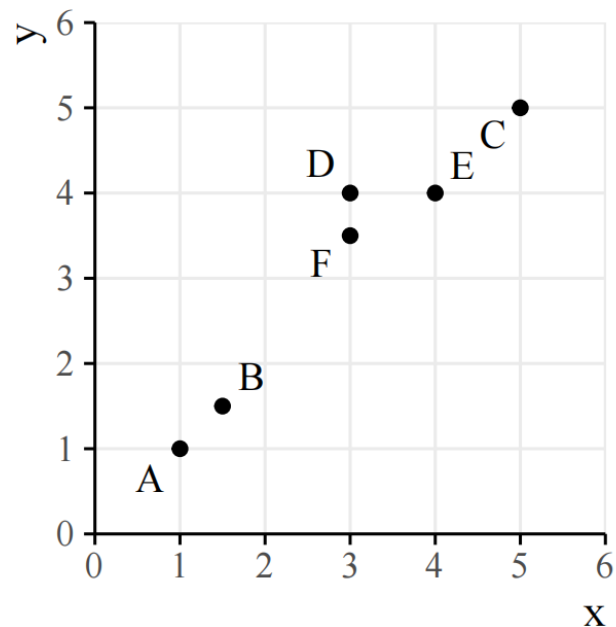


Exercise 8

1. Explain the difference between divisive and agglomerative hierarchical clustering.
2. Name the different inter-cluster distance measures (linkage method) and describe how they are computed.
3. Consider the dataset of 6 points given below. Perform an agglomerative clustering using Single/MIN linkage with Euclidean distance and create the dendrogram.

	x	y
A	1.0	1.0
B	1.5	1.5
C	5.0	5.0
D	3.0	4.0
E	4.0	4.0
F	3.0	3.5



4. Compute the distance between clusters $C_1 = \{A, B\}$ and $C_2 = \{D, E, F\}$ using:
 - (a) Complete/MAX linkage
 - (b) Group Average
 - (c) Ward's method
5. W.r.t your computations in question 4: If a new point $Z = (2, 2)$ was added to C_1 , the distance between C_1 and C_2 as computed by MAX linkage will:
 - Remain the same
 - not remain the same